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Out of Sight, Out of Mind: Divestments and the Global Reallocation of Pollutive Assets

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1 Big picture

- **Question.** When large listed firms face pressure to decarbonize, do they actually reduce global emissions, or do they mostly move polluting assets outside the public spotlight?
- **Main setting.** The paper studies global publicly listed firms reporting to CDP over 2011–2021, with special attention to Climate Action 100+ focus firms, called large emitters.
- **Empirical event.** The 2015 Paris Agreement is used as the key event that intensified investor, stakeholder, and public pressure on the largest greenhouse gas emitters.
- **Main decomposition.** The paper decomposes year-to-year changes in Scope 1 and Scope 2 emissions into four channels: boundary changes through divestments and acquisitions, changes in output, changes in methodology, and a residual like-for-like carbon-intensity component.
- **Central fact.** Large emitters reduce reported emissions after Paris, while other public firms do not. However, the reduction is concentrated in boundary changes, especially divestments.
- **Scale.** The authors estimate that approximately 369 million metric tons of CO₂e are reallocated through divestments in the post-Paris period.
- **No broad operational decarbonization.** The paper finds little evidence that large emitters reduce emissions through lower output, methodology changes, or residual intensity improvements.
- **Investor pressure channel.** Large emitters report a marked post-Paris increase in investor-related climate risks, while the increase in regulatory risks is much less differential.
- **Buyer-side result.** Pollutive assets sold by large emitters often move to firms that are less visible to public investors and climate activists, frequently privately held or outside the seller’s home region.
- **Takeaway.** Investor pressure can reshape corporate boundaries without necessarily reducing real emissions. A firm can look cleaner by selling dirty assets, but global emissions need not fall if the assets continue operating elsewhere.

2 Incremental contribution of the paper

2.1 Contribution relative to the climate-finance literature

- Prior climate-finance work studies emissions, disclosure, climate risk pricing, divestment campaigns, and investor engagement.
- This paper adds a boundary-of-the-firm channel: firms can respond to climate pressure by changing ownership of pollutive assets rather than by changing production technology.
- The paper therefore moves the analysis from *firm-level reported emissions* to *global ownership reallocation of emissions-producing assets*.

Interpretation: The incremental contribution is not simply showing that large emitters reduce reported emissions. It shows why reported reductions can be misleading when they are achieved through asset sales rather than physical abatement.

2.2 Contribution relative to corporate carbon accounting

- Scope 1 and 2 emissions are usually interpreted as operational emissions of a reporting firm.
- The paper shows that Scope 1 and 2 emissions can fall mechanically when polluting assets leave the reporting boundary.

- The authors hand-collect information on buyer and asset locations to trace what happens to divested emissions after boundary changes.

Interpretation: The paper contributes a measurement innovation: it separates accounting boundary changes from like-for-like changes in emissions intensity.

2.3 Contribution relative to ESG engagement debates

- One view is that investor engagement with large public firms is a powerful decarbonization tool.
- Another concern is that divestment or public pressure may only push dirty assets into less transparent ownership structures.
- This paper provides global evidence in favor of the second concern: pressure on listed firms can relocate emissions rather than eliminate them.

Interpretation: The paper’s policy relevance is that investor engagement must monitor real asset outcomes, not only the seller’s post-divestment emissions profile.

2.4 Contribution relative to merger and divestiture research

- The corporate finance literature often studies divestitures as value-enhancing refocusing or restructuring decisions.
- This paper studies divestitures as a response to nonfinancial stakeholder pressure and as a channel for reallocating environmental externalities.
- It shows that the identity of the buyer matters: public pressure is weaker if assets move to less visible or less climate-sensitive owners.

Interpretation: The paper extends the economics of divestitures from asset efficiency and valuation to environmental accountability and global externalities.

3 Data and measurement

3.1 CDP firm-level emissions data

- The main emissions data come from CDP and cover the 2011–2021 period.
- The baseline panel includes 1,354 listed firms across 45 countries.
- These firms account for almost 4 billion metric tons of annual carbon emissions.
- The key outcomes are gross Scope 1 and Scope 2 emissions.
- Scope 1 emissions are direct emissions from owned or controlled sources.
- Scope 2 emissions are indirect emissions from purchased electricity, heat, steam, or cooling.

Interpretation: CDP gives annual reporting data that allow the authors to observe both total emissions and self-reported reasons for changes in emissions.

3.2 Large emitters and control firms

- Large emitters are Climate Action 100+ focus firms.

- These firms are targeted by an investor-led initiative meant to pressure large global emitters to reduce emissions.
- The treatment group includes 111 such large emitters in the final sample.
- Control firms are other listed firms reporting to CDP.
- The authors compare large emitters to control firms in the same industry and region before and after 2015.

Interpretation: Climate Action 100+ status proxies for heightened investor, stakeholder, and public pressure after the Paris Agreement.

3.3 Emission-change decomposition

For firm i in year t , the paper decomposes changes in emissions into categories reported in CDP:

$$\Delta Emissions_{it} = Boundary_{it} + Output_{it} + Methodology_{it} + Residual_{it}. \quad (1)$$

- **Boundary** captures divestments and acquisitions.
- **Output** captures changes in production scale.
- **Methodology** captures changes in measurement or reporting methods.
- **Residual** captures like-for-like carbon-intensity changes not explained by the other categories.

Interpretation: The residual is the closest measure of real operational decarbonization. Boundary changes can make the firm look cleaner even if the physical emissions continue under another owner.

3.4 Divested asset data

- The authors hand-collect information on divested assets from firm filings, press releases, and other sources.
- They record seller identity, buyer identity, asset location, buyer public status, and buyer country.
- They distinguish asset sales from spin-offs.
- They calculate the emissions associated with divestments and aggregate them by seller and buyer regions.

Interpretation: This hand-collected data set is what allows the paper to go beyond reported emissions and evaluate whether dirty assets are reallocated globally.

3.5 Investor pressure and regulatory pressure

- Investor pressure is measured using CDP questions on perceived climate-related risks, especially investor-related risks.
- The authors also use textual analysis of annual reports to identify investor-related and environmental-regulation-related risk mentions.
- Regulatory pressure is proxied by country-level environmental stringency, particularly the WEF Stringency of Environmental Regulation score.

Interpretation: The paper separates investor pressure from regulatory pressure. The key divestment effects appear linked to investor-related risk exposure rather than simply tougher regulation.

4 Empirical specifications

4.1 Baseline difference-in-differences

The baseline firm-year specification is:

$$y_{it} = \beta LargeEmitter_i \times Post2015_t + \alpha_i + \alpha_{industry,t} + \alpha_{country,t} + Z'_i \gamma \times Post2015_t + \varepsilon_{it}. \quad (2)$$

- y_{it} is an emissions outcome or divestment outcome.
- $LargeEmitter_i$ equals one for Climate Action 100+ focus firms.
- $Post2015_t$ equals one after the Paris Agreement.
- Firm fixed effects absorb time-invariant firm differences.
- Industry-by-year and country-by-year fixed effects absorb sectoral and geographic shocks.
- Controls include pre-event average leverage and profitability interacted with $Post2015_t$.

Interpretation: The coefficient β measures whether large emitters change more after Paris than otherwise similar CDP-reporting firms in the same industry and country-year environment.

4.2 Dynamic event-study specification

The dynamic specification replaces $Post2015_t$ with year indicators:

$$y_{it} = \sum_{k=2010}^{2021} \beta_k LargeEmitter_i \times Year_{k,t} + FE + Controls + \varepsilon_{it}. \quad (3)$$

Interpretation: This specification checks whether large emitters and controls follow parallel trends before Paris and whether the treatment effect emerges after 2015.

4.3 Matched-sample specification

The paper also constructs matched control firms based on:

- country;
- industry;
- emission intensity.

The matched specification replaces industry-by-year and country-by-year fixed effects with match-cohort-by-year fixed effects.

Interpretation: Matching helps address the concern that large emitters are mechanically different from smaller CDP firms because they are concentrated in emission-intensive sectors or countries.

4.4 Risk-channel specifications

To test the investor-pressure mechanism, the paper estimates versions of the baseline model where:

- the dependent variable is whether the firm reports investor-related risk;
- the dependent variable is whether the firm reports regulatory, physical, or other climate risks;
- the large-emitter treatment is interacted with high investor-risk exposure.

Interpretation: The mechanism test asks whether the divestment response is stronger exactly where investor pressure becomes salient.

4.5 Deal-level asset-sale specifications

For divested assets, the paper estimates deal-level regressions of the form:

$$DealOutcome_d = \beta LargeEmitterSeller_d \times Post2015_d + FE + Controls + \varepsilon_d. \quad (4)$$

- Outcomes include whether the buyer is international, whether the asset is international, whether the buyer is publicly listed, and whether buyer environmental regulation is lower than seller regulation.
- The sample is based on divestment deals linked to emission-reducing boundary changes.

Interpretation: These regressions identify where divested pollutive assets go and whether they move out of the regulatory or investor spotlight.

5 Identification and credibility

5.1 Treatment variation

- Treatment is exposure to increased investor and public pressure after the Paris Agreement, proxied by Climate Action 100+ focus status.
- The comparison group consists of other listed CDP-reporting firms.
- Identification is difference-in-differences: large emitters are compared to controls before and after 2015.

Interpretation: The design is not randomized. Its credibility relies on controls, fixed effects, matching, and dynamic evidence that pre-trends are not driving the results.

5.2 Key threats and responses

- **Industry composition:** controlled with industry-by-year fixed effects and exact/nearest-neighbor matching.
- **Country policy shocks:** controlled with country-by-year or region-by-year fixed effects and environmental regulation measures.
- **Emission intensity differences:** addressed through emission-intensity matching and controls.
- **Plant closures rather than asset sales:** addressed using CDP text fields and hand-collected divestment data.
- **Broader trend of large firms shedding assets:** addressed with SDC and hand-collected deal-level tests.
- **Buyer-seller investor overlap:** addressed by examining crossholdings in buyers and sellers.

Interpretation: The paper’s case is strongest because the same pattern appears in reported emissions, reported reasons for emissions changes, hand-collected asset transactions, and buyer characteristics.

5.3 Limitations

- CDP data are self-reported, though the paper conducts robustness checks using verified or assured emissions.
- Treatment firms are larger and more visible than controls by construction.

- Divestment data are hand-collected and may miss small or opaque transactions.
- The paper documents reallocation, not the exact post-sale emissions trajectory of every asset under the buyer.

Interpretation: The main limitation is that the paper cannot observe every asset's actual emissions after the sale. The contribution is to show that ownership reallocation is large and systematic.

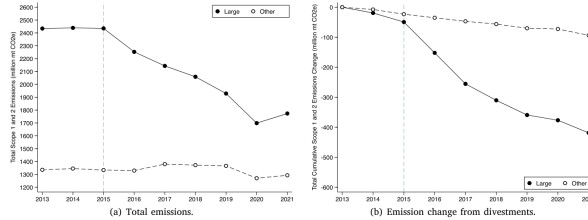
6 Tables and figures

6.1 Figures 1–2 and Tables 1–2: motivation and sample construction

Read:

- Figure 1 shows that after Paris, large emitters reduce total Scope 1 and 2 emissions, while other emitters do not. The same figure shows a large cumulative increase in emission reductions attributed to divestments.
- Figure 2 illustrates how CDP asks firms to explain the reason for changes in emissions.
- Table 1 describes the full firm-year sample, including firm characteristics, emissions, and emission-change decomposition.
- Table 2 describes the sample of large emitters by assets, revenue, emissions, geography, and industry.

Economic message: These visuals establish the empirical object: large emitters report falling emissions, and CDP allows the authors to identify whether this is due to real reductions or boundary changes.



1. Emissions and divestments around the 2015 Paris Agreement.
1 (a) of this figure shows the total combined gross Scope 1 and 2 emissions for a balanced sample of 613 firms around the 2015 Paris Agreement, split by large emitters ($N = 73$) and other emitters ($N = 540$). Panel (b) of this figure shows the combined cumulative gross Scope 1 and 2 emissions (in million tonnes CO₂e) that firms divested since 2013 (base year) for the same sample of firms.

Figure 1: emissions and divestments around Paris

Table 1

Descriptive statistics.

	Mean	Median	Std. Dev	Obs.
A. Firm characteristics				
Large emitter (0/1)	0.09	0.00	0.29	12,353
Total assets ('000 million USD)	24.44	9.02	43.01	12,353
Total revenue ('000 million USD)	15.85	6.48	25.96	12,353
Europe (0/1)	0.37	0.00	0.48	12,353
North America (0/1)	0.28	0.00	0.45	12,353
Rest of the world (0/1)	0.35	0.00	0.48	12,353
B. Emission data				
Scope 1 ('000 metric tonnes CO ₂ e)	3497.82	146.03	10,641.61	12,340
Scope 2 ('000 metric tonnes CO ₂ e)	693.76	185.43	1428.05	11,868
Scope 1+2 Growth Rate (%)	1.82	-1.01	25.29	10,785
C. Emission breakdown				
<i>Firm boundary</i>				
Firm boundary net (%)	0.46	0.00	5.24	12,331
Divestment (0/1)	0.12	0.00	0.32	12,353
M&A (0/1)	0.14	0.00	0.35	12,353
<i>Output</i>				
Output net (%)	1.38	0.00	7.32	12,331
Decreased output (0/1)	0.17	0.00	0.37	12,353
Increased output (0/1)	0.32	0.00	0.47	12,353
<i>Method</i>				
Method net (%)	0.63	0.00	5.67	12,331
Decreased method (0/1)	0.11	0.00	0.32	12,353
Increased method (0/1)	0.15	0.00	0.36	12,353
<i>Residual</i>				
Residual net (%)	-1.39	-1.77	21.45	10,785
Residual < -1% (0/1)	0.55	1.00	0.50	10,785
Residual < -5% (0/1)	0.32	0.00	0.47	10,785
Residual > 1% (0/1)	0.27	0.00	0.44	10,785
Residual > 5% (0/1)	0.16	0.00	0.37	10,785

This table reports descriptive statistics at the firm- (reporting) year level. The sample period is 2011 to 2021.

Table 1: full sample descriptive statistics

Reason	Direction of change	Emissions value (percentage)	Please explain calculation
Change in renewable energy consumption	Decreased	2.99	*1)
Other emission reduction activities	Decreased	7.39	*2)
Divestment	Decreased	6.08	*3)
Acquisitions			
Mergers			
Change in output	Increased	3.37	*4)
Change in methodology			
Change in boundary			
Change in physical operating conditions			
Unidentified			
Other	Increased	0.12	*5)

*1) "In Electricity Generation, a) the increase in renewable capacity of 73MW in wind capacity in Spain in 2021 resulted in 45,755 tCO₂e avoided [...] b) the increase in renewable capacity of 206MW in wind capacity and 101MW in solar PV capacity in Chile in 2021 resulted in 177,501 tCO₂e avoided [...]"

*2) "Regarding Scope 1, a) in Electricity Generation the shut-down of all coal power plants in Spain implies a reduction of 1,067,906 tCO₂e in Electricity Generation as a result of the Energy Efficiency Operational Plan (E.E.O.P.), resulting in a reduction in specific fuel consumptions, [...] which implies the reduction of 73,788 tCO₂e. Regarding Scope 2, c) in Electricity Distribution, the decrease in electricity losses in transport and distribution in Spain implies a reduction of 5 tCO₂e [...]"

*3) "The divestment in electricity networks in Chile implies a reduction of 939,057 tCO₂e (scopes 1 and 2). [...] the percentage of emission decrease can be calculated as 939,057 tCO₂e / 15,455,482 tCO₂e = 6.08%."

*4) "a) The increase in LNG activities implies an increase of 359,712 tCO₂e; b) the increase in natural gas distribution activities imply an increase of 119,597 tCO₂e; c) the increase in electricity distribution activities implies an increase of 41,943 tCO₂e; [...] the percentage of emission decrease can be calculated as 321,252 tCO₂e / 15,455,482 tCO₂e = 3.37%."

*5) "Increase in emissions due to different small factors. The sum of all of them accounts for an increase in emissions of 18,583 tCO₂e (0.12% of total emissions in 2020)."

CDP Information on changes in gross global Scope 1 and 2 emissions.
are shown the response by Natogy Energy Group SA to question C7.9a ("Identify the reasons for any change in your gross global emissions (See 1), and for each of them specify how your emissions compare to the previous year") for the 2022 CDP survey wave (which asks about information on 2021 emissions). Emissions are in '000 tonnes CO₂e. Methods used: Scope 1 and 2 emissions for the year 2021 were 1,479,906 tCO₂e.

Figure 2: CDP emission-change categories

Descriptive statistics – Large emitters.

Panel A. Firm characteristics (averages over 2011 to 2015 period)				
	Mean	Median	Std. Dev	Obs.
Total assets ('000 million USD)	92.33	66.45	74.47	111
Total revenue ('000 million USD)	59.62	46.35	47.80	111
Scope 1 ('000 metric tonnes CO ₂ e)	23,460.00	12,429.31	25,201.66	111
Scope 2 ('000 metric tonnes CO ₂ e)	2719.39	1594.59	2789.60	108
Panel B. Geographical distribution		Panel C. Industries (2-digit SIC)		#
<i>North America (N = 41)</i>		Electric and Gas services (49)		26
USA		Petroleum refining (29)		15
Canada		Transportation equipment (37)		15
		Chemicals (28)		9
<i>Europe (N = 46)</i>		Oil and Gas extraction (13)		6
Austria		Metal mining (10)		5
Denmark		Food and Kindred products (20)		5
Finland		Stone, Clay, Glass, and Concrete (32)		5
France		Primary metal (33)		5
Germany		Transportation by air (45)		4
Ireland		Public sector (Other) (99)		4
Italy		Other		13
Luxembourg				
Netherlands				
Norway				
Spain				
Switzerland				
United Kingdom				
<i>Other Countries (N = 24)</i>				
Australia				
Brazil				
Colombia				
Japan				
Mexico				
Russia				
South Africa				
South Korea				
Taiwan				
Thailand				

This table reports descriptive statistics for the sample of 111 large CO₂ emitters (see main text for details).

of Scope 1 emissions within the country; industries show [...] of smaller firms with lower absolute emission levels [...]

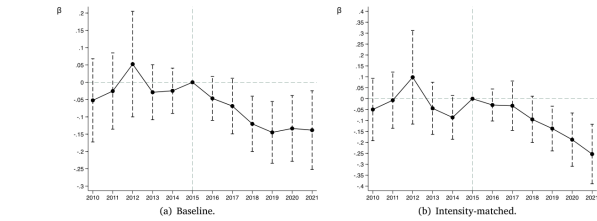
Table 2: large emitter descriptive statistics

6.2 Figure 3 and Table 3: emission reductions

Read:

- Figure 3 shows the dynamic effect of large-emitter status on emissions around the Paris Agreement.
- The post-2015 coefficients for Scope 1 emissions and emissions intensity become negative.
- Table 3 shows that large emitters reduce Scope 1 emissions and emissions intensity relative to controls after Paris.
- The effects survive fixed effects and intensity-matched samples.

Economic message: Large emitters look cleaner after Paris, but this table does not yet show whether the reduction is real abatement or a boundary change.



dynamic effect on firm emissions.

re examines the combined Scope 1 and 2 emissions of large emitters around the 2015 Paris Agreement. Results are reported for two sample panels and (ii) emission-intensity matching. See the main text for details on the matching procedures. Estimated coefficients are from

Figure 3: dynamic effect on emissions

Table 3
Changes in firm emissions.

Sample:	Base				Intensity-matched			
	Restriction:				High Em industries			
Variable	ln(Scope 1) (1)	ln(Scope 1) (2)	ln(Scope 2) (3)	Δ Intensity (4)	ln(Scope 1) (5)	ln(Scope 1) (6)	ln(Scope 2) (7)	Δ Intensity (8)
Large Em, $\times$ Post,	-0.162*** (0.005)	-0.137*** (0.036)	-0.016 (0.844)	-0.040*** (0.001)	-0.139*** (0.028)	-0.121* (0.095)	0.056 (0.636)	-0.054*** (0.002)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,331	12,172	11,660	11,700	6250	2707	2589	2416
Adj R ²	0.96	0.96	0.93	0.11	0.96	0.96	0.96	0.99

This table examines the emission activity of large emitters around the 2015 Paris Agreement. The unit of observation is the firm-year level i . The sample period is 2011 to 2021. Large Em equals one if firm i is a large CO2 emitter (defined in more detail in the main text), and zero otherwise. Post is an indicator variable that is equal to one for firm reporting years after 2015, and zero otherwise. Scope is the gross global emission (in metric tonnes CO2e) of firm i in reporting year t . Columns 1-2 and 5-6 report results for Scope 1 emissions; columns 3 and 7 report results for Scope 2 emissions. In column 5 the sample is restricted to industries with an above median (pre-Paris) emission intensity Δ Intensity is the change in Scope 1 emission intensity of firm i from reporting year $t-1$ to t . Emission intensity is defined as gross global Scope 1 emission (1000 metric tonnes CO2e) scaled by firm revenue (million USD). Results are reported for two samples: (i) the full baseline panel and (ii) emission-intensity matching. In the emission-intensity matching sample, treated firms are matched to control firms with similar emission intensity within the same country and industry. See the main text for details. The regressions include firm fixed effects, year fixed effects, year $\times$ industry (2-digit SIC code) fixed effects, and industry fixed effects. Δ Intensity is the change in Scope 1 emission intensity of firm i from reporting year $t-1$ to t . Emission intensity is defined as gross global Scope 1 emission (1000 metric tonnes CO2e) scaled by firm revenue (million USD). Results are reported for two samples: (i) the full baseline panel and (ii) emission-intensity matching. In the emission-intensity matching sample, treated firms are matched to control firms with similar emission intensity within the same country and industry. See the main text for details. The regressions include firm fixed effects, year fixed effects, year $\times$ industry (2-digit SIC code) fixed effects, and industry fixed effects.

Table 3: changes in emissions

6.3 Tables 4–5 and Figure 4: reasons for reported emission changes

Read:

- Table 4 shows that large emitters are more likely to report divestments as the reason for emission reductions.
- Other categories, such as output changes, method changes, and residual intensity changes, do not provide the main explanation.
- Figure 4 shows a post-Paris increase in divestment activity among large emitters.
- Table 5 decomposes intensive and extensive margins and confirms that the central divestment effect is at the extensive margin.

Economic message: The post-Paris emission reduction comes mostly from changing firm boundaries, not from operational decarbonization.

Table 4
Reasons for changes in firm emissions.

Panel A. Activities that decrease emissions

Sample:	Base		Matched		Base		Matched		Base		Matched	
Variable:	Divestments (0/1)		Output (0/1)		Method (0/1)		Residual < - 5% (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)				
Large Em. × Post,	0.089*** (0.005)	0.106*** (0.005)	0.037 (0.216)	0.003 (0.944)	-0.030 (0.241)	-0.080** (0.011)	-0.080** (0.027)	-0.017 (0.713)				
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
Industry × Year	Yes	–	Yes	–	Yes	–	Yes	–				
Country × Year	Yes	–	Yes	–	Yes	–	Yes	–				
Cohort × Year	–	Yes	–	Yes	–	Yes	–	Yes				
Observations	12,193	2711	12,193	2711	12,193	2711	10,599	2377				
Adj R ²	0.25	0.31	0.20	0.14	0.10	0.10	0.08	-0.03				
Mean of dep var:	0.12	0.16	0.17	0.21	0.11	0.13	0.32	0.29				

Panel B. Activities that increase emissions

Sample:	Base	Matched	Base	Matched	Base	Matched	Base	Matched
Variable:	MBA (0/1)		Output (0/1)		Method (0/1)		Residual > 5% (0/1)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Large Em. × Post,	0.016 (0.319)	0.004 (0.894)	-0.038 (0.313)	-0.062 (0.181)	-0.051* (0.069)	-0.061* (0.066)	-0.014 (0.576)	-0.025 (0.453)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,193	2711	12,193	2711	12,193	2711	10,599	2377
Adj R ²	0.19	0.13	0.22	0.11	0.13	0.12	0.05	-0.02
Mean of dep var:	0.14	0.13	0.32	0.38	0.15	0.15	0.16	0.14

This table examines the channels through which emission activities of large emitters change around the 2015 Paris Agreement. The unit of observation is the firm-year level i . The sample period is 2011–2021. Large Em equals one if firm i is a large CO₂ emitter (defined in more detail in the main text), and zero otherwise. Post is an indicator variable equal to one for firm reporting years after 2015, and zero otherwise. The dependent variables are indicator variables equal to one if firm i reports that its combined global Scope 1 and 2 emissions changed in year t due to: (i) divestments, (ii) output changes, (iii) changes in emissions measurement (Method), (iv) merger activity, or (v) other reasons, including efficiency improvements and emission-reduction activities (Residual). Panel A (B) examines activities that reduced (increased) emissions. Results are reported for two samples: (i) the full baseline panel and (ii) emission-intensity matching. In the emission-intensity-matched sample, treated firms are matched to control firms with similar emission intensity within the same country and industry. See the main text for details. The regressions include firm fixed effects, year $\times$ industry (2-digit SIC code) fixed effects, year $\times$ country fixed effects, and industry fixed effects.

Table 4: reasons for emission reductions

Table 5
Reasons for changes in firm emissions – Ex- and intensive margin.

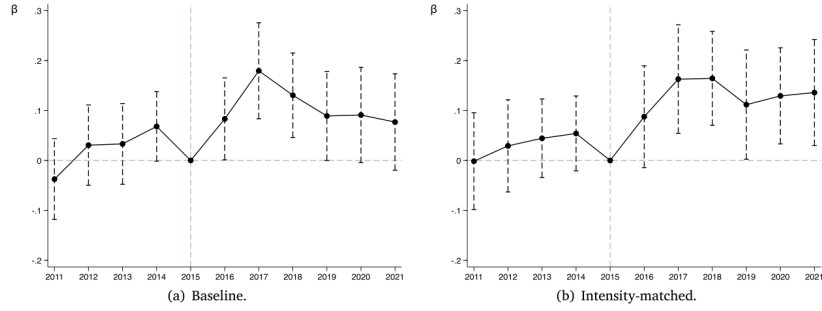
Panel A. Intensive margin

Sample	Base		Matched		Base		Matched		Base		Matched	
	Boundary net %		Output net %		Method net %		Residual net %					
Variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)				
Large Em, $\times$ Post,	-0.89 (0.756)	-10.97* (0.056)	1.01 (0.325)	2.45* (0.085)	-1.88 (0.625)	8.16 (0.434)	-0.59 (0.623)	-1.64 (0.240)				
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
Country $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
Cohort $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
Observations	2127	213	5070	1257	2514	240	10,560	2347				
Adj R ²	0.10	-0.24	0.18	-0.03	0.11	-0.61	0.02	-0.09				
Mean dep var:	2.90	1.00	3.55	2.57	3.70	-0.27	-1.39	-1.40				

Panel B. Ex- and intensive margin

Sample:	Base		Matched		Base		Matched		Base		Matched	
	Boundary net %		Output net %		Method net %		Residual net %					
Variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)				
Large Em, $\times$ Post,	-2.33** (0.029)	-4.44*** (0.005)	0.16 (0.825)	0.98 (0.363)	0.06 (0.914)	2.71** (0.013)	-0.06 (0.978)	-7.08*** (0.008)				
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,799	2408	10,799	2408	10,799	2408	10,599	2377				
Adj R ²	0.28	0.58	0.41	0.25	0.12	-0.01	0.41	0.21				
Mean dep var:	0.48	0.06	1.46	1.46	0.63	-0.11	-1.70	-1.78				

Table 5: ex- and intensive margins



dynamic effect on firm divestment activities. We examine the divestment activity of large emitters around the 2015 Paris Agreement. Results are reported for two samples: (i) the baseline and (ii) emission-intensity matching. See the main text for details on the matching procedures. Estimated coefficients are from the following regression:

6.4 Figure 5, Table 6, Figure 6, Figure 7, and Table 7: investor pressure channel

Read:

- Figure 5 decomposes annual emissions changes into divestments, M&A, output, method, and residual components for large and other emitters.
- Table 6 shows that large emitters report more investor-related climate risks after Paris, while the evidence is weaker for physical or regulatory risks.
- Figure 6 shows a dynamic increase in investor-related risk mentions after Paris.
- Figure 7 uses annual reports and shows that investor-related risk mentions among large emitters rise from about 3% in 2013 to close to 50% in 2021.
- Table 7 shows that emission reductions and divestment activity are stronger among firms reporting higher investor-related risk.

Economic message: The mechanism evidence points toward investor pressure, not only environmental regulation, as the driver of divestment-based decarbonization.

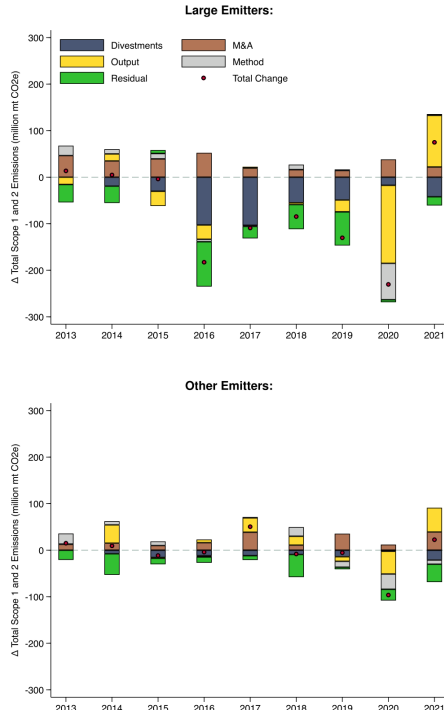


Figure 5: annual emission changes by category

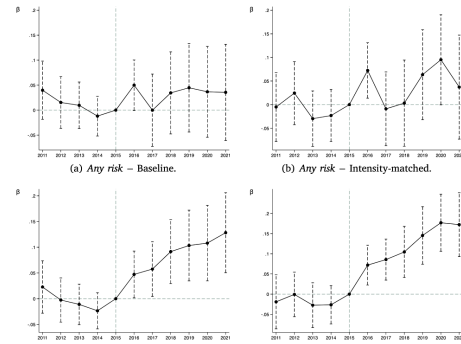


Figure 6: dynamic investor-related risk

Table 7
Emissions and divestment activity – Investor risk.

Sample:	Panel A. ln(Scope 1)				Panel B. Divestments (0/1)			
	Base		Intensity-matched		Base		Intensity-matched	
Investor risk:	Any	High	Any	High	Any	High	Any	High
Large Em, $\times$ Post, $\times$ (D = 1) _i	-0.164*	-0.238**	-0.166	-0.325***	0.142***	0.175***	0.221***	0.229***
	(0.092)	(0.038)	(0.202)	(0.005)	(0.007)	(0.002)	(0.000)	(0.001)
Large Em, $\times$ Post, $\times$ (D = 0) _i	-0.102	-0.081	-0.094	-0.036	0.055	0.054	0.037	0.054
	(0.147)	(0.230)	(0.270)	(0.686)	(0.138)	(0.145)	(0.402)	(0.214)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	–	–	Yes	Yes	–	–
Country $\times$ Year FE	Yes	Yes	–	–	Yes	Yes	–	–
Cohort $\times$ Year FE	–	–	Yes	Yes	–	–	Yes	Yes
Observations	12,172	12,172	2707	2707	12,193	12,193	2711	2711
Adj R ²	0.96	0.96	0.98	0.98	0.25	0.25	0.30	0.30
Coeff. equal? (p-val)	0.59	0.22	0.64	0.05**	0.15	0.06*	0.02**	0.03**

This table examines the emission and divestment activity of large emitters around the 2015 Paris Agreement. The unit of observation is the firm-year level it . The sample period is 2011 to 2021. In columns indicated with *Any*, $D = 1$ ($D = 0$) is an indicator variable that is equal to one if the firm reports *any* (*no*) investor-related risk exposure, and zero otherwise. In columns indicated with *High*, $D = 1$ ($D = 0$) is an indicator variable that is equal to one if the firm reports investor-related risk exposure with a score of *medium-high* [4–6] (*low to medium* [1–3]), and zero otherwise. See Table 6 for details. All other variables are defined in Tables 3 and 4. Panel A (B) examines emission (divestment) activity. Results are reported for two samples: (i) the full baseline panel and (ii) emission-intensity matching. See Table 3 and the main text for details on the matching procedures. The regressions include firm fixed effects, year $\times$ industry (2-digit SIC code) fixed effects, year $\times$ country fixed effects, and match-cohort $\times$ year fixed effects, when indicated. Further controls include firm-level characteristics (pre-event average leverage and profitability interacted with *Post*). p -values based on robust standard errors, clustered at the firm level, are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

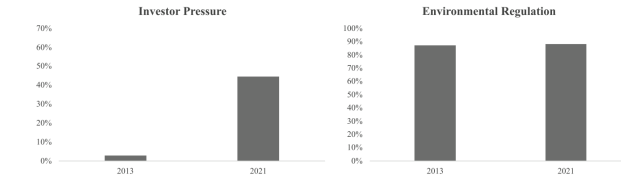
Table 6
Risk factors.

Panel A. Dependent variable: ln(Risk score [1–6])								
Variable:	Base		Base		Base		Base	
	Physical	Matched	Regulatory	Matched	Investor	Matched	Other	Matched
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Large Em, $\times$ Post,	-0.06 (0.250)	-0.12* (0.069)	0.05 (0.232)	0.03 (0.601)	0.16** (0.040)	0.15* (0.056)	-0.09 (0.194)	-0.21*** (0.003)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls $\times$ Post FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Post FE	Yes	–	Yes	–	Yes	–	Yes	–
Country $\times$ Post FE	Yes	–	Yes	–	Yes	–	Yes	–
Cohort $\times$ Post FE	–	Yes	–	Yes	–	Yes	–	Yes
Observations	2654	584	2654	584	2654	584	2654	584
Adj R ²	0.49	0.27	0.55	0.45	0.54	0.21	0.41	0.10
Mean of dep. var.:	1.32	1.31	1.37	1.39	0.30	0.29	1.20	1.23

Panel B. Dependent variable: Any risk (0/1)								
Variable:	Base		Base		Base		Base	
	Physical	Matched	Regulatory	Matched	Investor	Matched	Other	Matched
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Large Em, $\times$ Post,	-0.03 (0.206)	-0.07* (0.053)	0.03 (0.230)	-0.03 (0.244)	0.12** (0.023)	0.13** (0.018)	-0.07 (0.137)	-0.19*** (0.000)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls $\times$ Post FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Post FE	Yes	–	Yes	–	Yes	–	Yes	–
Country $\times$ Post FE	Yes	–	Yes	–	Yes	–	Yes	–
Cohort $\times$ Post FE	–	Yes	–	Yes	–	Yes	–	Yes
Observations	2654	584	2654	584	2654	584	2654	584
Adj R ²	0.39	0.09	0.40	0.03	0.53	0.22	0.32	-0.00
Mean of dep. var.:	0.90	0.91	0.93	0.95	0.22	0.22	0.77	0.82

This table explores risk factors that large emitters report around the 2015 Paris Agreement. The unit of observation is the firm-period level it , i.e., the sample period is split into a pre-Paris period (2011 to 2015) and a post-Paris period (2016 to 2021). The dependent variables measure whether firms are exposed to (i) physical risks, (ii) regulatory risks, (iii) investor-related risks, or (iv) other risks (categories are defined in more detail in the main text). Panel A uses ln(Risk Score) as dependent variable, where risk scores range from 1 (no risk) to 6 (high risk). Scores are defined as maximum scores by period and category. Panel B uses indicator variables that are equal to one if the firm reports any risk exposure in the respective category and period, and zero otherwise. All other variables are defined in Table 3. Results are reported for two samples: (i) the full baseline panel and (ii) the emission-intensity-matched sample. Treated firms are matched to control firms with similar emission intensity within the same country and industry. See the main text for details. The regressions include firm fixed effects, *Post* $\times$ industry (3-digit SIC code) fixed effects, *Post* $\times$ country fixed effects, and match-cohort $\times$ *Post* fixed effects, when indicated.

Table 6: climate-risk reporting



Investor and environmental regulation risks in annual reports.

Figure shows the fraction of large emitters that mention risks related to investor ESG preferences (left) or risks related to environmental issues (right).

Figure 7: annual-report risk mentions

6.5 Table 8: regulatory pressure robustness

Read:

- Table 8 controls for cross-country environmental regulation stringency.
- The large-emitter post-Paris emission and divestment effects remain economically and statistically meaningful.
- The SER interactions do not explain the main divestment result.

Economic message: The main effect is not simply that large emitters are located in countries with stricter environmental regulation.

Table 8
Stringency of environmental regulation.

Panel A. Emission activity – Dependent variable: $\ln(\text{Scope } 1)$						
Sample:	Baseline			Intensity-matched		
	(1)	(2)	(3)	(4)	(5)	(6)
Large Em, $\times$ Post,	-0.127** (0.038)	-0.144** (0.022)	-0.133** (0.034)	-0.121* (0.095)	-0.125* (0.099)	-0.129* (0.076)
SER Score, $\times$ Post,		-0.085** (0.025)	-0.063 (0.109)		0.084 (0.347)	0.002 (0.981)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	–	Yes	Yes
Country $\times$ Year FE	Yes	–	–	–	–	–
Region $\times$ Year FE	–	–	Yes	–	–	Yes
Cohort $\times$ Year FE	–	–	–	Yes	–	–
Observations	12,172	12,244	12,244	2707	2783	2783
Adj R^2	0.96	0.96	0.96	0.98	0.98	0.98

Panel B. Divestment activity – Dependent variable: Divestment (0/1)						
Sample:	Baseline			Intensity-matched		
	(1)	(2)	(3)	(4)	(5)	(6)
Large Em, $\times$ Post,	0.089*** (0.005)	0.087*** (0.008)	0.089*** (0.007)	0.106*** (0.005)	0.109*** (0.002)	0.107*** (0.003)
SER Score, $\times$ Post,		0.014 (0.191)	0.012 (0.317)		0.016 (0.729)	0.018 (0.706)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	–	Yes	Yes
Country $\times$ Year FE	Yes	–	–	–	–	–
Region $\times$ Year FE	–	–	Yes	–	–	Yes
Cohort $\times$ Year FE	–	–	–	Yes	–	–
Observations	12,193	12,266	12,266	2711	2786	2786
Adj R^2	0.25	0.25	0.25	0.30	0.34	0.35

This table examines the emission (Panel A) and divestment (Panel B) activity of large emitters around the 2015 Paris Agreement. The unit of observation is the firm-year level it . The sample period is 2011 to 2021. *Scope 1* is the global Scope 1 emissions (in mt CO₂e) of firm i in reporting year t . *Divestments* is an indicator equal to one if firm i indicates that their emissions changed in year t because of divestments. *Large Em* equals one if firm i is a large CO₂ emitter (defined in more detail in the main text), and zero otherwise. *Post* is an indicator variable that is equal to one for firm reporting years after 2015, and zero otherwise. *SER Score* is the stringency of environmental regulation (SER) in the firm's headquarter country. Results are reported for two samples: (i) the full baseline panel and (ii) the emission-intensity-matched sample.

6.6 Tables 9–12 and Figure 8: where divested assets go

Read:

- Table 9 shows that large emitters are more likely after Paris to complete divestment deals and sell international assets.
- Table 10 confirms the same broad patterns in a smaller hand-collected deal sample.
- Table 11 shows that European sellers account for a large increase in sold emissions after Paris, while buyers are often outside Europe.
- Table 12 shows that divestment effects are strongest for high-emission-intensity firms and for emissions-reducing divestments.
- Figure 8 shows limited investor overlap between sellers and buyers. A large share of divested emissions goes to private buyers or public buyers without seller-buyer crossholdings.

Economic message: Pollutive assets tend to move away from the public and investor spotlight. The result is not just internal reshuffling among the same shareholders.

Table 9

Deal-level analysis.

Panel A. Deal location and buyer public status						
Variable:	Base			Intensity-matched		
	Int Deal (1)	Int Asset (2)	Public (3)	Int Deal (4)	Int Asset (5)	Public (6)
Large Em, $\times$ Post,	0.095*** (0.007)	0.128*** (0.003)	-0.079** (0.014)	0.106** (0.048)	0.034 (0.575)	-0.020 (0.685)
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6025	6025	6025	884	884	884
Adj R^2	0.19	0.15	0.09	0.27	0.19	0.15

Panel B. Regulatory stringency and investor pressure						
Variable:	Base			Intensity-matched		
	Lower SER (1)	Lower CL100 (2)	Lower SER- CL100 (3)	Lower SER (4)	Lower CL100 (5)	Lower SER- CL100 (6)
Large Em, $\times$ Post,	0.095*** (0.008)	0.055 (0.136)	0.078** (0.043)	0.089** (0.041)	0.027 (0.554)	0.096** (0.041)
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5847	4600	5847	859	713	859
Adj R^2	0.19	0.34	0.28	0.28	0.41	0.31

This table examines the divestment activity of large emitters around the 2015 Paris Agreement. The unit of observation is at the deal level. The sample period spans 2011 to 2021. *Large Em* equals one if the seller is a large CO2 emitter (defined in more detail in the main text), and zero otherwise. *Post* is an indicator variable that is equal to one for deals completed after 2015, and zero otherwise. Divestment information is from SDC Platinum. For treatment firms, the sample is restricted to deals completed in years in which the firm reports divestments that reduced Scope 1 and 2 emissions. *Int Deal* is an indicator equal to one if the buyer is headquartered in a different country than the seller, and zero otherwise. *Int Asset* is an indicator equal to one if the divested asset is not located in the seller's headquarters country, and zero otherwise. *Public* is an indicator equal to one if the buyer is a publicly listed non-financial firm, and zero otherwise. *Lower SER* is an indicator equal to one if the stringency of environmental regulation (SER) score is lower in the buyer country versus the seller country. *Lower CL100* is an indicator equal to one if the CL100+ Ratio is lower in the buyer country versus the seller country. *Lower SER-CL100* is based on *Lower CL100* but missing values for the CL100+ Ratio in the buyer and seller country are predicted using the countries SER score (cf. Figure A-3, Panel A). Results are reported for two samples: (i) the full baseline panel and (ii) the emission-intensity-matched sample. Treated firms are matched

Table 9: deal-level evidence

Table 11

Divestment activity by large emitters – Seller and buyer region.

Panel A. Asset sales										
Pre Paris Agreement:										
	Volume (mil mt CO2e)					# Deals				
	Total	EU	NA	RoW	Unk.	Total	EU	NA	RoW	Unk.
Sell	112.1	81.2	30.0	0.9	—	101	74.0	25.0	2.0	—
Buy	112.1	34.7	37.0	39.2	1.1	101	21.0	43.5	35.5	1.0
Net (B-S)	—	-46.5	7.0	38.3	1.1	—	-53.0	18.5	33.5	1.0
Post Paris Agreement:										
	Volume (mil mt CO2e)					# Deals				
	Total	EU	NA	RoW	Unk.	Total	EU	NA	RoW	Unk.
Sell	267.1	176.3	74.1	16.7	—	141	92.0	42.0	7.0	—
Buy	267.1	44.1	115.8	104.7	2.6	141	28.5	47.8	57.7	7.0
Net (B-S)	—	-132.3	41.6	88.0	2.6	—	-63.5	5.8	50.7	7.0
Post-Pre Paris Agreement:										
	Volume (mil mt CO2e)					# Deals				
	Total	EU	NA	RoW	Unk.	Total	EU	NA	RoW	Unk.
Net (Po-Pr)	155	-85.7	34.6	49.6	1.5	40	-10.5	-12.7	17.2	6.0
Panel B. Spin-offs										
	Volume (mil mt CO2e)					# Deals				
	Total	EU	NA	RoW	Unk.	Total	EU	NA	RoW	Unk.
Pre Paris	25.5	4.7	20.8	—	—	7	2.0	5.0	—	—
Post Paris	90.8	82.2	8.2	0.4	—	6	3.0	2.0	1.0	—
Post-Pre	65.3	77.5	-12.7	0.4	—	-1	1.0	-3.0	1.0	—

This table reports descriptive statistics on the divestment activities of large emitters. We collect information on divestment activities for all firm-years in which a large emitter reports that divestments reduced their total combined Scope 1 and 2 emissions by at least 1% (139 firm-years). We report separate statistics for the period before the Paris Agreement (2010-2015) and the period after the Paris Agreement (2016-2021). Panel A reports information on all sales of assets and subsidiaries. "Volume sold" is the total Scope 1 and 2 emission reduction in million mt CO2e that results from divestments by large emitters that are headquartered in region j = (Europe (EU), North America (NA), Rest of the World (RoW), Unknown (Unk.)). "Volume bought" is the flipside of "volume sold" and identifies in which regions the divested emissions end up (based on the headquarters location of the buyer(s)). That is, for each period and across all regions, total volume sold = total volume bought. When identifying buyers, we take into account that sellers can engage in multiple divestments per year that involve different buyers (potentially located in different regions). As CDP information on emission reductions due to divestments is only available at the (selling-) firm-year level, we approximate emissions at the deal $\times$ buyer level as follows: First, we split the total reported emission reduction due to divestments by seller i in year t equally across all deals of the seller that we can identify in the respective year. Second, in case a deal involves multiple buyers, we equally split emissions across all firms. The number of deals (# Deals) is defined in the same manner (e.g., for a deal with two buyers, each buyer is assigned a deal share of 0.5). Panel B reports information on spin-off transactions. For spin-offs we do not differentiate between buyer and seller region, as the spun-off entity remains a stand-alone company (whose shares are offered to the current shareholders of the seller).

Table 11: seller and buyer regions

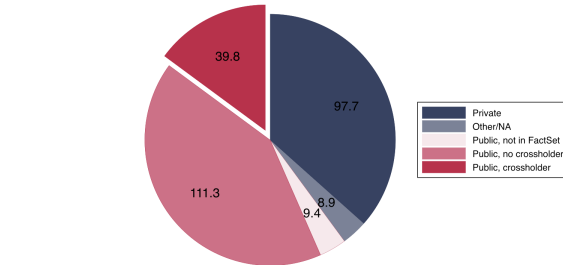


Fig. 8. Investors in buyers of divested assets: Crossholdings.

This figure provides information on the investor structure of buyers of divested assets after the Paris Agreement (see Table 11 in the main paper for details on the sample). "Crossholders" are investors that hold shares in both buyers and sellers of assets. The sample is restricted to the post Paris Agreement period and distinguishes between private and public (i.e., listed) buyers. The figure indicates the deal emission volume (mil mt CO2e) for each category.

Variable:	Int Deal (1)	Int Asset (2)	Public (3)	Lower SER (4)	Lower CL100 (5)	Lower SER-CL1 (6)
Post 2015,	0.109** (0.032)	0.076 (0.443)	-0.011 (0.912)	0.051 (0.546)	0.081* (0.069)	0.106** (0.032)
Country	Yes	Yes	Yes	Yes	Yes	Yes
Observations	234	234	242	233	170	233
Adj R^2	0.23	0.27	-0.02	0.25	0.53	0.42

This table examines the divestment activity of large emitters around the 2015 Paris Agreement. The unit of observation is at the deal level. The sample period spans 2011 to 2021. *Post* is an indicator variable that is equal to one for deals completed after 2015 and zero otherwise. This table uses hand-collected divestment information for large emitters as described in Table 11. All variables are defined in Table 9. The regressions include seller-country fixed effects, when indicated. p -values based on robust standard errors, clustered at the firm level, are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

Table 10: hand-collected sample

Table 12

Robustness: Divestment activity of large firms.

Panel A. Emission-reducing divestment activity				
Emission intensity:	Low	Medium	High	
Variable:	Any Divest (0/1) (1)	Any Divest (0/1) (2)	Non-Reduced Divest (0/1) (3)	Emission-Reduced Divest (0/1) (4)
Large Firm, $\times$ Post,	-0.044 (0.119)	0.009 (0.824)	0.103*** (0.005)	
FEs and controls	Yes	Yes	Yes	
Observations	4715	3885	3458	
Adj R^2	0.18	0.22	0.33	

Panel B. Overall divestment activity				
Emission intensity:	Low	Medium	High	
Variable:	Any Divest (0/1) (1)	Any Divest (0/1) (2)	Non-Reduced Divest (0/1) (3)	Emission-Reduced Divest (0/1) (4)
Large Firm, $\times$ Post,	-0.016 (0.659)	0.038 (0.396)	-0.051 (0.204)	
FEs and controls	Yes	Yes	Yes	
Observations	4715	3885	3458	
Adj R^2	0.31	0.33	0.33	

Panel C. International divestment activity				
Emission intensity:	Low	Medium	High	
Variable:	Any Divest (0/1) (1)	Any Divest (0/1) (2)	Non-Reduced Divest (0/1) (3)	Emission-Reduced Divest (0/1) (4)
Large Firm, $\times$ Post,	0.016 (0.603)	0.047 (0.225)	-0.074** (0.033)	
FEs and controls	Yes	Yes	Yes	
Observations	4715	3885	3458	
Adj R^2	0.28	0.35	0.26	

Table 12: robustness by emission intensity

7 Short summary

- The paper studies how large global emitters reduce reported Scope 1 and 2 emissions after the Paris Agreement.
- Large emitters reduce reported emissions much more than other listed CDP-reporting firms.
- The reduction is mainly attributed to divestments and other boundary changes, not to residual like-for-like improvements.
- Large emitters report rising investor-related climate risks after Paris.
- The divestment response is stronger among firms that report investor-related risks.
- Regulatory pressure does not fully explain the effect.
- Hand-collected asset-sale data show that pollutive assets are reallocated globally.
- Many buyers are private firms or public firms without substantial crossholding overlap with sellers.
- The paper estimates that 369 million metric tons of CO₂e are reallocated through divestments after Paris.
- The central lesson is that firm-level emissions disclosures can improve while global emissions remain unchanged if dirty assets continue operating under less visible owners.

8 Conclusion

8.1 Core conclusion

Berg, Ma, and Streitz show that investor pressure after the Paris Agreement induced large public emitters to reduce reported emissions mostly by selling pollutive assets. The assets often moved to firms that were less exposed to investor scrutiny, implying that public-market climate pressure can reallocate emissions rather than eliminate them.

8.2 Economic intuition

The firm facing investor pressure internalizes the reputational or financing cost of having emissions on its own books. Selling the asset removes those emissions from the seller's reported footprint. However, the buyer may face weaker pressure and may continue operating the asset. The private incentive is to move emissions out of sight; the social objective is to reduce emissions globally.

8.3 Policy implication

The paper implies that climate disclosure and investor engagement must track assets, not only firms. Otherwise, capital-market pressure can create appearance-based decarbonization by shifting dirty assets into private or less scrutinized ownership.